

CHAPTER 2

The Physiology of a Better Engine

The human body is an incredible machine. You take in food, an all-encompassing term that covers everything from fresh berries to a Big Mac, and within minutes convert it (or at least some of it) into usable energy. When it comes to accessing that energy, you are able to go from the sedentary state of reading this book to sprinting, if necessary, at a moment's notice. And as you run longer or change your pace, your body seamlessly adjusts how it produces energy based on how quickly you are demanding it and the energy sources available.

To do this, the human body has three primary energy systems that produce adenosine triphosphate (ATP), which is the energy currency the body uses to power all activities. The quickest source is the immediate energy system of stored ATP and creatine phosphate (ATP-PCr), followed by the glycolytic (anaerobic) system and the aerobic system. All three energy systems produce ATP, which releases energy when one of its three phosphate bonds is broken. The resulting adenosine diphosphate (ADP) is then resynthesized to ATP so it can be broken down again, and again, and again. All three energy systems are always working together synergistically in varying amounts (Figure 2.1); there is no on/off switch, and at any given time the amount of energy produced by each system is based on demand. These three energy systems are fundamental to endurance training, and although you don't need a physiology degree to be a good ultrarunner, it is nonetheless helpful to understand the systems you're training.

The Immediate Energy System: ATP-PCr

The ATP-PCr system supports high-power efforts that last less than eight to ten seconds. You use it when you have to jump out of the way of a speeding bus, and from an athletic standpoint it's most important in power sports like sprinting, mixed martial arts, or powerlifting. In endurance running this system is used mostly for explosive movements like jumping across a creek or bounding up a series of boulders. During those few seconds, you demand energy faster than either the glycolytic or aerobic energy system can deliver it. The ATP-PCr system yields immediate energy because ATP, the energy-yielding molecule, is stored directly in the muscle. This very limited supply that is stored in your muscles is available for immediate use because it doesn't require the multiple steps that the anaerobic and aerobic systems require to produce ATP. Endurance athletes don't rely heavily on this system, and it is typically adequately developed through normal training.

The Lactic Acid System

The lactic acid system is the high-speed express lane for delivering energy quickly in times of heightened demand. This is the system people often refer to as "anaerobic," which literally means "without oxygen." This terminology causes confusion because it implies that the body has stopped using oxygen to produce energy, which is not the case. As exercise intensity increases, there comes a point at which the demand for energy outstrips the aerobic system's ability to produce ATP in working muscles. Then comes a hill. Or you spot a fellow racer and pick up the pace. The demand for energy increases further, and in order to meet the demand, the mitochondria's aerobic functions are supported by energy derived from anaerobic processes. Although the actual process involves many steps, to put it simply, where aerobic metabolism delivers all of the energy available from a molecule of glucose (sugar), glycolysis converts glucose into lactate through a process that delivers some of the energy available in glucose, but more quickly.

Lactate is a normal by-product in the breakdown of carbohydrate (glucose) to ATP during glycolysis. As lactate is produced via glycolysis, 75 percent of it enters the bloodstream and is shuttled to the mitochondria, where it can be utilized as a substrate for energy production (Brooks 1986). The remaining 25 percent is sent to the liver to be repurposed into glucose via a process called “gluconeogenesis,” literally “creating new glucose.” Lactate has been misunderstood for years. It has been blamed for the burning sensation in your muscles when you surge above your sustainable pace. It has been blamed for delayed-onset muscle soreness. People have tried to massage it away, flush it out, and buffer it in (largely failed) attempts to enhance or prolong exercise. We now understand that increases in lactate are not the cause of fatigue, they merely coincide with it, and the best way to utilize lactate is to reintegrate it back into normal aerobic metabolism to complete the process of breaking it down into energy, water, and carbon dioxide.

Under normal conditions, when aerobic energy supply meets energy demand, lactate clearance from the blood occurs at a rate faster than it accumulates. Once lactate production outstrips the body’s ability to buffer it, lactate begins to accumulate in the blood. This is referred to as the “lactate threshold” and indicates a change in the way energy is being derived from predominantly aerobic means to now include a greater contribution from anaerobic sources. In short, this threshold is the intensity at which an athlete can sustain his effort for an extended period of time. By increasing the intensity or pace at which this occurs, an athlete is able to improve performance. Thus, this is one of the primary outcomes of endurance training. This training adaptation also enables you to recover from hard efforts more quickly, because deriving energy from glycolysis is like buying energy on credit. You’re getting the currency you need as you need it, but you don’t have unlimited credit, and sooner rather than later you’re going to have to pay back every cent you borrowed. What’s more, you have to cut back on spending while you’re paying it back, which means you have to slow down.

As an endurance athlete, one of the key adaptations is an improvement in the ability to integrate lactate into aerobic energy production so it can be oxidized completely. The faster you can process lactate, the more work you can perform before fatigue-inducing anaerobic metabolites (not lactate) accumulate in the muscle and blood. Or, in the financial analogy, a stronger aerobic system puts more cash (aerobic metabolism) in your pocket so you’re not so quick to use credit.

The Aerobic System

The aerobic system is remarkable. It produces the majority of the energy we use for daily life and exercise; can burn carbohydrate, fat, and protein simultaneously; and can regulate the mixture it burns based on fuel availability and energy demand. It can even burn protein, although that’s not used as a significant source of energy unless you have very low energy reserves. The aerobic system is a flex-fuel engine that’s clean and efficient; when the aerobic system is done with a molecule of carbohydrate, the only waste products are water and carbon dioxide. In comparison, the lactic acid system discussed earlier produces energy faster but can only utilize carbohydrate, produces less ATP from every molecule it processes, and produces metabolites that lower body pH and interfere with muscle contractions.

For an endurance athlete, increasing the amount of oxygen the body can transport and utilize is one of the primary goals of training, because oxygen is the limiting factor in how quickly you can produce energy aerobically. On the transportation side, your body’s amount of hemoglobin, which is a component of your blood, largely dictates how much oxygen you can take from your lungs and deliver to working muscles. Once oxygen has been transported to the muscles, it is then utilized by the mitochondria during the process of energy conversion. The mitochondria are a muscle cell’s power plants. They harness the oxygen and fuel (carbohydrate and fat) and convert those substrates into useable energy. Training improves both the transportation and utilization sides of the equation.

VO2 Max

When the body is delivering and utilizing as much oxygen as it possibly can and exercise intensity reaches its absolute peak, you're at VO2 max. VO2 max, or maximum aerobic capacity, is measured in liters of oxygen per minute taken up and utilized by the muscles. Since VO2 max varies by body size and muscle mass, the measurement unit mL/kg/min is used to compare across different athletes because it takes body weight into account.

VO2 max is one of the indicators (but not the only indicator) of your potential as an endurance athlete. And while a higher VO2 max is generally a good thing for any individual athlete, an exceedingly high VO2 max doesn't automatically guarantee you'll become a champion. All it means is you have a big engine. You need a big engine to be an elite athlete, but no matter what size engine you start with, you can optimize your performance with effective training.

It takes a great effort to reach intensities near VO2 max, and VO2 max-specific workouts generate an enormous amount of lactate and burn energy tremendously fast. But the reward is worth the effort, because increasing your pace at VO2 max (the pace you can run while utilizing the maximum amount of oxygen possible) and your absolute VO2 max (how many liters of oxygen you can utilize per minute) give you the physiological underpinnings to run faster at any distance. Ultramarathon athletes sometimes suffer from a one-pace mentality. When your goal is to keep moving for fifteen to thirty hours, it seems to make the most sense to primarily use endurance and some lactate threshold workouts to get to the fitness level necessary to sustain that effort. It's the sustainable aspect that keeps too many athletes from venturing into more intense efforts. The perception is that the sport is all about making steady forward progress and that one only needs longer long runs to ensure success. While volume is important, intensities at and near your VO2 max are also critical for building your engine.

The Endurance String Theory

Delineating the various ways your body can produce energy is both a blessing and a curse. On the positive side, knowing how each system works gives us the information necessary to design training that maximizes the specific physiological adaptations we desire. On the downside, the same information has inadvertently led people to believe that these systems operate independently of each other. Sports scientists, coaches, and even the folks who made your heart rate monitor have told you that training in "zone something-or-other" will target your lactic acid system and increase your pace at lactate threshold. And although that is true, the lactic acid system isn't the only one doing the work at that intensity, nor is it the only one that will reap a training benefit.

Energy production is always coming through all possible pathways, but your demand for energy determines the relative contribution from each. At low to moderate intensities, such as 40 to 50 percent of VO2 max or an easy endurance run, the vast majority of your energy comes from the aerobic system (mitochondria breaking down primarily fat and some carbohydrate). As the intensity level increases, a greater amount of energy is derived anaerobically. Energy contribution from glycolysis ramps up even further as you approach intensities near and above your lactate threshold, which can be anywhere from 65 to 85 percent of VO2 max for a particular athlete. Because glycolysis uses only glucose, the overall percentage of energy coming from carbohydrate rises dramatically as intensity increases from lactate threshold to VO2 max. The body is still using fat, but at a lower rate because the breakdown of fat into energy is much slower.

Rather than seeing your various energy pathways as separate and distinct, it's better to think of them as segments of one continuous string, arranged based on the relative contributions of energy derived from each as the duration of an all-out task increases. At one end of the spectrum, for very short efforts lasting eight to ten seconds, is a small segment represented by energy contributions primarily from the ATP-PCr system. After that is the lactic acid system, which can provide a lot of energy at full gas for efforts lasting less than about two minutes. Finally, as duration increases further, there is a large segment representing the aerobic system, which theoretically could power your muscles at a moderate intensity level forever if it had sufficient oxygen and fuel. Improving fitness in one system is like lifting the string in that region—all other areas of the string rise, too. The extents of these ancillary improvements vary based on the system you initially targeted. For instance, targeting VO2 max has a greater lifting effect on lactate-threshold fitness and aerobic endurance than training at aerobic intensities has on lifting lactate threshold or VO2 max. Ultimately, all of the systems are interconnected, and even the most narrowly focused training will have upstream and downstream effects on your physiology.

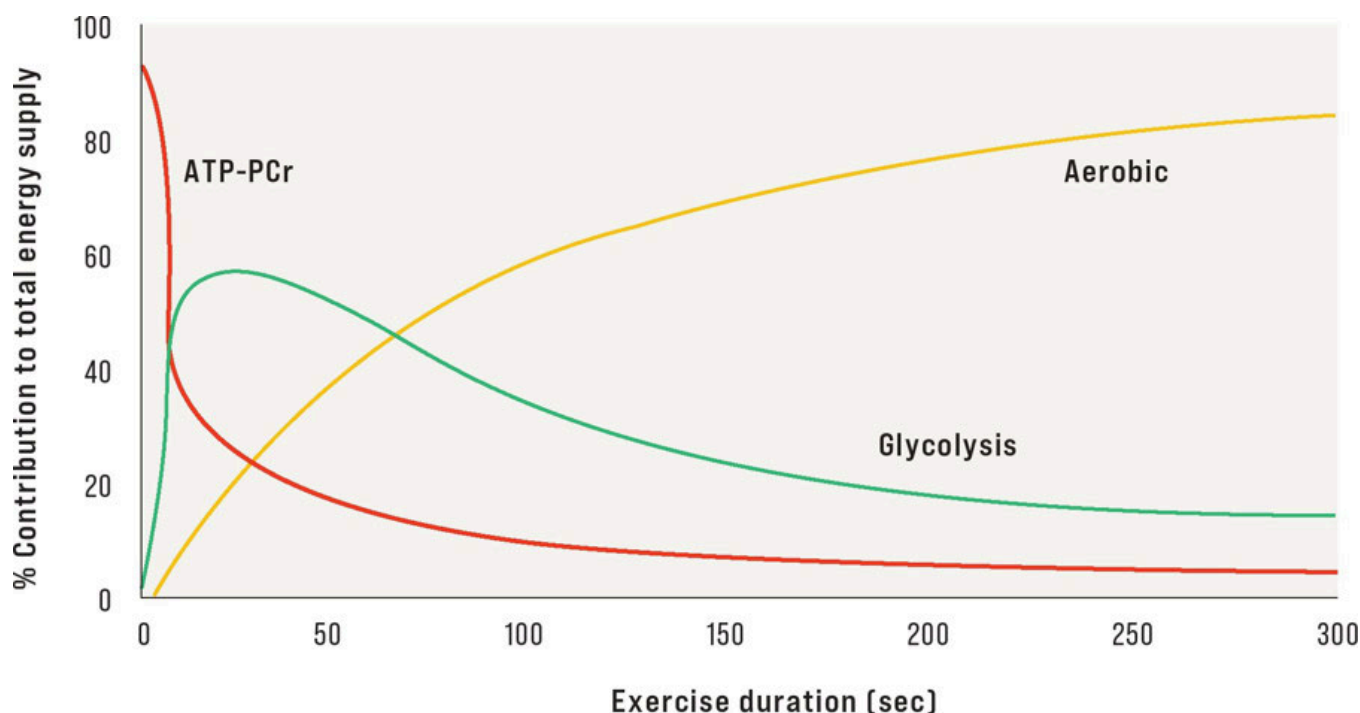


FIGURE 2.1 The contributions of the three energy systems (ATP-PCr, glycolysis, and aerobic) to energy production over time (seconds) during an all-out effort. Adapted from Gastin 2001.

WHAT ABOUT AEROBIC DEFICIENCY SYNDROME?

THE TERM “AEROBIC DEFICIENCY SYNDROME” (ADS) was coined by Phil Maffetone as a deleterious condition whereby your aerobic system is “deficient” (Maffetone 2015). He has described it as a “devastating” condition for athletes that affects “millions of people.” The result of this syndrome, according to Maffetone, is “reduced endurance,” “quicken fatigue,” “loss of aerobic speed,” and “a variety of chronic conditions.” Sound pretty terrible? Yeah, I’d be afraid of it, too.

This apparent syndrome has also been regurgitated among writers, bloggers, coaches, athletes, and anyone else that can’t think for themselves as a justification for avoiding high-intensity intervals and only performing very low-intensity exercise.

Let’s examine these claims.

First, as of this book’s publishing, the term “aerobic deficiency syndrome” appears nowhere in any scientific literature. It’s not in any textbook or scientific journal article. Contrast that with overtraining syndrome (oftentimes referred to as underperformance syndrome), which has been extensively described in scientific literature and has a robust consensus among athletes, coaches, and sports scientists.

Furthermore, as the name implies, people “suffering” from ADS apparently have a deficient aerobic system stemming from too much high-intensity work. The fact of the matter is, after you get past just a minute or two of an all-out effort, nearly 100 percent of the energy contributions will come from an aerobic pathway (see Figure 2.1). Thus, for any endurance athlete, even if you are doing high-intensity interval work, you are still deriving almost all of that energy aerobically. So, the notion that your “aerobic

system is deficient” is complete and total nonsense. Aerobic deficiency syndrome is just a fictitious combination of words, is not described anywhere in any scientific literature, and does not accurately describe anything in endurance training.

Bottom line, this is just a made-up term. Don't be fearmongered into not doing high-intensity work because of a fake syndrome.

Fundamental Principles of Training

No two athletes are exactly the same, but we all use the same mechanisms to produce energy. Similarly, no two athletes adapt to training the exact same way, but effective training is based on a common set of principles. When you distill the world's most successful training programs across all sports, you arrive at five distinct principles of training:

1. OVERLOAD AND RECOVERY

2. PROGRESSION

3. INDIVIDUALITY

4. SPECIFICITY

5. SYSTEMATIC APPROACH

OVERLOAD AND RECOVERY

Fundamentally, training is designed to leverage the body's normal response to stress (or overload). The human body is designed to respond to overload, and as long as you properly overload a system and allow time for recovery, that system will adapt and be ready for the same or greater stress in the future. To achieve positive training effects, this principle must be applied both to individual training sessions and to entire periods of your training. For instance, an interval workout around your lactate threshold must be intense enough and long enough so that the total stimulus is great enough to make your body say, “OK, I got it. I need to be bigger, stronger, and faster than I am right now so that the next time this happens, I am more equipped for the task.”

Proper overload and subsequent recovery are incredibly important for ultrarunners. Most ultrarunners will come into the sport with some running background. The significance of this is that they have previously gone through the early stages of development where improvement and PRs seemingly happen every week. Runners with more than four years of experience (most ultrarunners fit into this category) will know that as you gain experience, improvement is harder and harder to come by. Striving for those gains means finding ways to increase overload, often within limited training time, while also scheduling adequate recovery. As a result, training and workout architecture will matter more for an experienced athlete than for a novice athlete who can improve simply by not

screwing it up. The various workouts, short-range plan, and long-range plan described later in this book are designed so that you can realize an appropriate amount of overload and then combine that with appropriate recovery in order to continue to see improvement regardless of your experience level.

Many novice athletes start out with haphazard or scattered training, but they nevertheless make steady gains because they are beginners. Just the act of training leads to significant improvement when you're starting out. But that progress stalls relatively quickly because you reach the point where the stimulus applied with each individual workout is not high enough or consistent enough to lead to further adaptation. Focusing your training on one area for a number of weeks, as you can do with a block of lactate threshold training, concentrates workload and training time to create a stimulus large enough to improve performance in that area. This becomes even more important for ultra-endurance athletes, because you are already adapted to a high overall workload. To make progress in any one aspect of fitness, you not only need to focus on it but also need to reduce focus on other areas during the same period. For example, when working on VO2 max it's important to reduce overall volume to adequately recover between sessions and maintain the quality of each workout.

On the other end of the spectrum, there's recovery. Recovery, which is covered in depth in Chapter 8, is not merely the absence of workouts but rather a crucial component of training. Days off should not be viewed as missed opportunities to get in another run. In reality, the periods between your workouts are when the really important adaptations such as plasma volume expansion, mitochondrial biogenesis, and red blood cell volume increases happen in your body. When you're in the middle of a training run, you're not improving your fitness; you're just applying stress and accumulating fatigue. But when you kick up your feet, sleep, hydrate, and provide your body with adequate and proper nutrition, that's when your fitness improves. It's not possible to train hard and gain fitness without proper rest and recovery. So, the next time your Type A buddy who's trained every day of the past four years chastises you for sitting on the couch or going for a walk with the dog instead of a sixty-minute run, just smile and tell her you're busy adapting.

HOW LONG SHOULD INTERVALS BE?

POP QUIZ: EACH OF THE INTERVAL SETS BELOW features the exact same duration of work, but only one is the most effective for improving VO2 max. Which one?

A. 12 x 1 minute at maximum intensity with 1 minute recovery between intervals

B. 4 x 3 minutes at maximum intensity with 3 minutes recovery between intervals

C. 2 x 6 minutes at maximum intensity with 6 minutes recovery between intervals

Intervals that work are long enough to stress the desired physiology and separated by recovery periods that let an athlete repeat the effort. Intervals that don't work are either too short or too long and separated by recovery periods that are too short or too long. They might be hard but don't have a focus in a way that is productive. You get tired but ultimately not stronger or faster.

The point of intervals is to maximize the body's exposure to intensity over time. For intervals where the primary purpose is to improve VO2 max, they need to be long enough to achieve and sustain an effort above 90 percent of your VO2 max (as measured by oxygen consumption), which takes anywhere from sixty to ninety seconds. The figure below is from a study that examined the

physiological response to one-, two-, four-, and six-minute intervals with 1:1 work-rest ratios. Though each combination represents twenty-four minutes of interval work, you can see the differences in cardiovascular response as the intervals get longer. They found that athletes achieve their highest oxygen consumption during the four-minute intervals. In practice, intervals between two and four minutes are long enough for an athlete to ramp up to VO2 max intensity and maintain it for the duration of the interval. When athletes pace themselves through six-minute intervals, they don't reach as high an intensity.

OPTIMAL WORK-REST RATIOS

The interval duration is one thing to get right; the work-rest ratio is another. For intervals targeting VO2 max improvement, two to four minutes at a 1:1 work-rest ratio works best, as it maximizes recovery from the previous interval without extending the overall workout duration and thus compromising the intensity you can achieve during the rest of the interval set. As intervals get longer and the intensity drops, athletes don't need as much recovery relative to the interval length before they are ready to repeat the effort. That's why the work-rest ratio for intervals at lactate threshold (TempoRun) is 2:1 and grows even greater for high-end aerobic intervals (SteadyStateRun). (TempoRuns and SteadyStateRuns will be further defined in Chapter 9.)

Getting back to the original question on the most effective VO2 max workout, the answer is B (4 x 3 minutes). The intervals in A are too short to reach VO2 max and stay there long enough to be effective. The intervals in C are so long most athletes would pace themselves at too low an intensity to optimally train VO2 max.

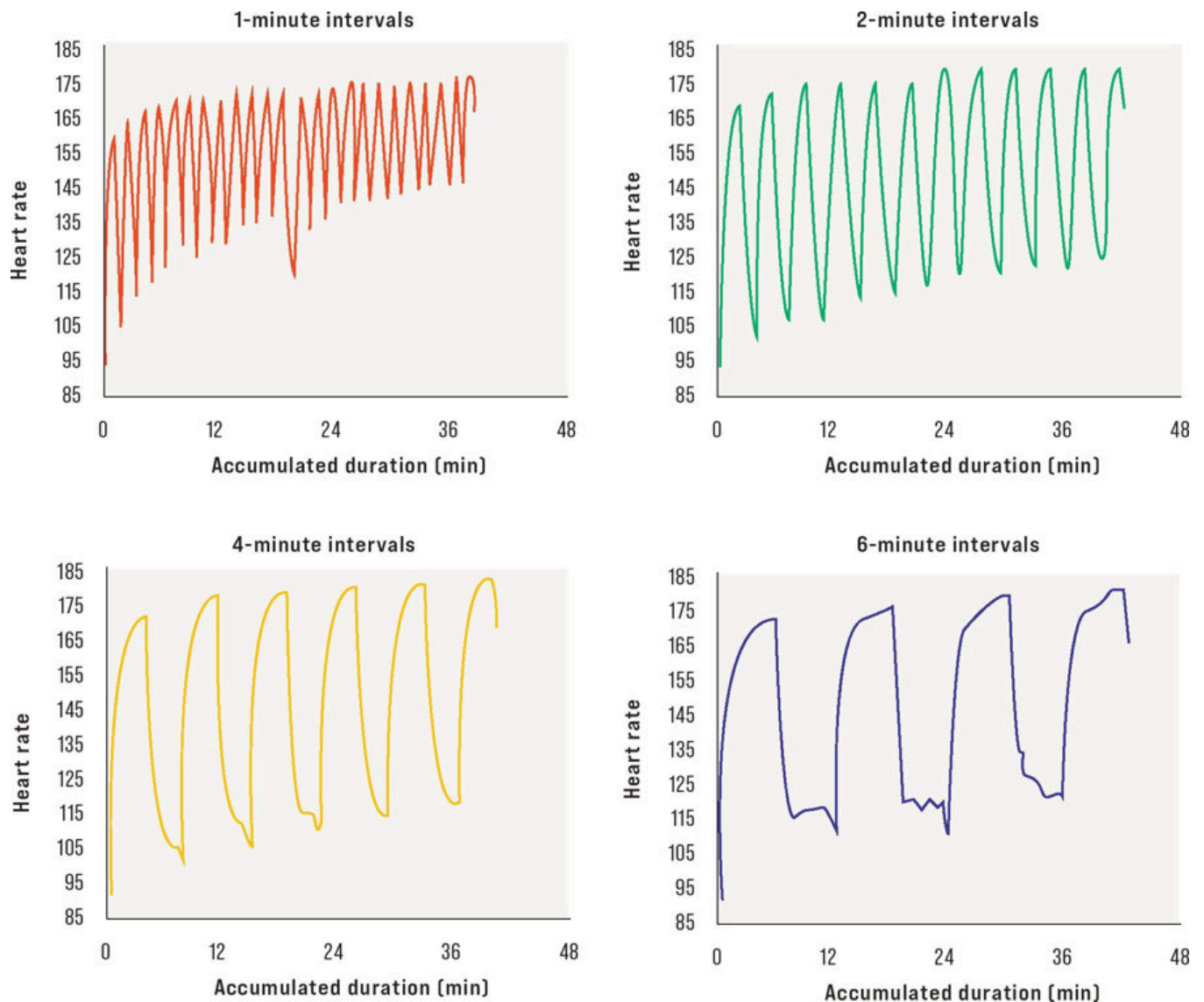


FIGURE 2.2 Mean heart rate responses during (A) 1-minute, (B) 2-minute, (C) 4-minute, and (D) 6-minute intervals. Adapted from Seiler and Sjsuren 2004.

PROGRESSION

Training must progressively move forward for continued gains in performance to occur. Time and intensity are the two most significant variables you can use to increase your workload. For instance, you can increase the number of hours you devote to training (which is training volume) or increase the overall intensity of your workouts. You can use time and intensity to manipulate training in a near limitless combination. No matter what you choose, the end result must generate a training stimulus great enough to elicit adaptations from your heart, muscles, blood vessels, bones, gut, connective tissue—basically everything. These are not static adaptations, however. Once you adapt and grow stronger, you have to again manipulate the time and intensity variables to further increase workload and generate another training stimulus. In other words, it will take a bigger workload to overload a stronger system.

The timeline for progression can take several forms. Many novice marathon training programs will use a weekly progression structure, meaning the volume (the amount of time or miles you do) and volume of intensity (the amount of time spent doing hard efforts) increase every week with periodic recovery weeks. Practically speaking, these runners would do a thirty-minute Tempo workout one week, a forty-minute Tempo the next week, and a

forty-five-minute Tempo the week after that assuming that some sort of positive adaptation takes place in those intervening weeks that enables the athlete to handle higher and higher workloads.

This type of architecture, while fine for many athletes, has a few glaring flaws upon further examination. First, for most cardiovascular- and endurance-related adaptations, the time course of said adaptations takes weeks, not days, to complete. That being said, there are very few things going on physiologically that would result in any meaningful improvement between weeks one, two, and three. Second, the hardest Tempo workouts (forty-five minutes) would be happening during the week of the most cumulative fatigue, presenting a potential “worst of both worlds” scenario (highest level of fatigue coupled with the hardest workout). Finally, the athlete is set up to do their easiest hard workout (as oxymoronic as that sounds) at the time when they are the freshest (week one, presumably after a recovery week). It is for these reasons that I primarily prefer that volume and/or volume of intensity generally decrease throughout a phase, then increase (or progress) subsequently after a recovery week. Using the above example, the fortyfive-minute Tempo run would be on week one, forty-minute Tempo on week two, and thirty-minute Tempo on week three when the athlete has the most cumulative fatigue. In this way, progression can be viewed through the lens of a few or even several weeks at a time. This concept will be discussed more in Chapter 12.

Neither training time nor intensity is limitless, even for professional athletes. There are only twenty-four hours in the day, and the human body can only be pushed so hard. Professional athletes—across the range of endurance sports—are pretty much maxed out in terms of the annual hours they can accumulate while still performing at a high level. While some case studies have shown that simply adding low-intensity training volume for highly trained or elite athletes results in continued improvement (Solli et al. 2017), others contradict this, suggesting that for athletes who are already trained, adding more volume may not lead to additional improvements in VO2 max, power at lactate threshold, or mitochondrial density (Laursen et al. 2002).

Unlike professional athletes who can design their lives around training, most amateur athletes are limited in terms of the time they can devote to training. For almost everyone lining up at an ultramarathon, running is a priority that has to be balanced with—and almost always comes after—many other priorities. The other commitments in your life mean you have to do what you can in the time you have. To achieve progression within the weekly training hours you have available, you have to manipulate all of the tools you have available at some point or another. This means increasing the overall training time, types of intensity, and volume of intensity.

SEQUENCE OF TRAINING

PROGRESSION IS OFTEN CONFUSED WITH “FASTER” in the context of the sequence of training. In other words, as the year progresses, most athletes think that their training should progress to higher and higher intensities, which is common with runners training for events like the 5K, 10K, and marathon. In those disciplines, beginning with a slower “aerobic base” followed by higher intensities and ultimately culminating with intervals at a race-specific intensity is the norm. However, that does not mean that the sequence of training has to progress from lower-intensity aerobic training to medium-intensity training and finally to higher-intensity training. Rather, progression should be thought of as occurring within each intensity so that the system gets stronger and needs a bigger stimulus (volume and intensity) in order to adapt.

Starting off with low intensity and moving to higher intensities is a stereotypical— and flawed—way of organizing training, particularly for ultrarunners. As will be discussed in Chapter 10, an ultrarunner’s training does not have to move through this order of low to medium to high intensity. Rather, it should move from developing the least-specific aspects of your ultramarathon physiology to the most specific.

Individuality

The individuality principle simply states that the training program that works for you, right down to the individual workouts and interval intensities, has to be based on your physiological and personal needs. Training is not a one-size-fits-all product. All parts of your program—the total mileage, the number and type of intervals, and even the terrain—must be personalized. That doesn't mean that you can't train with your friends or training partners; it just means that while you're with them, you have to stay true to your own program.

The individuality principle is another reason there are no full-length training plans in this book. As I'll say many times throughout the book, ultrarunning is not just a longer marathon. If I were writing a marathon running or Olympic-distance triathlon training book, I'd include training plans, because a generalized training plan can work for events of those lengths. Whenever I have tried to write generalized training plans for ultrarunning events, they just don't work that well. A training plan for Western States 100-Mile Endurance Run, Hardrock, Wasatch Front 100 Mile Endurance Run, and similar races would have to be thirty weeks long, minimum. With a time frame that long, a prewritten training plan is bound to under- or overestimate an individual runner's response to training and her ability to stick to the schedule. It will end up being way too hard and will run you into the ground, or way too easy and will not adequately prepare you for the demands of the event. In addition, it's impossible for a static training plan to fully consider the milieu of other training considerations including nutrition, equipment, psychological preparedness, environmental considerations, pacing strategies, and the like.

Ideally, every athlete would work with a coach and get a training program built from scratch, but personal coaching is not an option for everyone. The workouts and concepts in this book are rooted in the principles I use to coach my athletes, and you'll be able to apply the individuality principle to these workouts as well when you establish your personal training program and fit the workouts into your busy work and family schedules.

Specificity

Your training must resemble the activity you want to perform. In a broad sense, this means that if you want to be a runner, you should spend the vast majority of your training time running. In a narrower sense, it means you have to determine the exact demands of the activity you wish to perform and tailor your training to address those demands, particularly as any critical races draw near. Conversely, it also means that your training is going to prepare you optimally for specific events and activities.

The importance of specificity becomes clear when you look at how competitive ultrarunning has evolved over the past decade. The speed at the front of the pack has increased dramatically, so much so that running a course record pace from a decade ago might land you well outside of the top ten now. Is this because today's runners are that much more talented than runners from ten years ago? Are they running more miles than athletes did ten years ago? Are the shoes that much better? Are the energy bars that much better? No, none of the above. Athletes at the top of the sport are getting faster because their training has become more specific to the unique demands of the events they are training for. And the more we have learned about tailoring training to those demands, the more athletes of all ability levels can benefit from the same knowledge.

Systematic Approach

When it comes to achieving high-performance fitness, you need a training program that integrates and addresses all the principles of training. A systematic approach to training integrates all the crucial components: overload and recovery, progression, individuality, and specificity. Focusing on any one of the principles while neglecting others will take your training off course.

Ultrarunners devote a lot of time to training, so it is unfortunate when I see athletes wasting much of that time with ineffective workouts (see the earlier sidebar "Sequence of Training") and poorly planned programs. Workouts that are neither hard enough to contribute to positive adaptations nor easy enough to provide adequate

recovery just contribute to fatigue. Scattered training plans that jump from one intensity to the next before anything has substantially developed make athletes work hard without creating meaningful progress. It's all bang, but very little buck.

In order to leverage the benefits of each of the previous four principles of training, they need to be combined into a systematic approach to improvement. Appropriate levels of overload and recovery must be established based on your individual needs and manipulated so that you achieve progression. And a training program doesn't do you much good unless it prepares you for the specific and unique demands of your goal event.

You can apply any of the previous four principles individually, but you're not likely to be satisfied with the results. A common failure of training is achieving mastery of overload and recovery and progression while completely neglecting individuality and specificity. I see this most often with the data junkies, the athletes so focused on numbers, graphs, and training logs that all they care about is the trend of the data, even if it's leading them away from the fitness they need to perform at their best in their goal event.

Another problematic scenario is created by neglecting individuality and progression. This is typically an issue for social runners, athletes who value the social environment of the running community so highly that they substitute socializing for progression and individuality, which causes their training progress to stall or even collapse. These athletes are essentially going through the motions or treading water. Preparing for an ultramarathon requires a lot of focus for a significant period of time, and that level of focus may not be sustainable year after year. Therefore, a balance between focused training and some social running is ideal for most runners. I would much rather see runners reduce their focus on individuality and progression for a while and focus on staying active in the running community than drop out or burn out completely. In the former situation, they feel less pressure to keep pushing their fitness upward but still experience the satisfaction of training and being an active athlete.

Some people seem to be able to remain fit regardless of what they do or how much they do. These people are the fortunate anomalies. Most athletes, even those of you who like to think of yourselves as rebels, thrive with structure and benefit significantly from approaching training systematically. As you plan your training and get ready for your ultramarathon, remember that with limited training time, every hour and every interval counts, and all workouts are connected through the principles of training.

Special Considerations for Female Athletes

AUTHOR'S NOTE: I am the first person to admit that I don't know everything. I am in a constant pursuit of diving down the next rabbit hole to figure out the best ways to improve athletes. In that endeavor, I rely on a combination of research and experience to form best practices. So, it should come as no surprise that when it comes to special considerations for female athletes, I am woefully underexperienced. Can I coach and advise female athletes? Absolutely. But I felt that this section of the book would be best penned by a female coach who can accurately and authentically combine areas of research and personal experience. Fortunately, I have just the person for the job in fellow coach and co-author of this book, Corrine Malcolm. The following section was written directly by Corrine. It follows a framework from another contributor to this book, Nick Tiller, PhD, whose original research can be found in the References section (Tiller et al. 2021).

Women's participation in endurance events has come a long way. In 1928, several women collapsed upon completing the Olympic 800m race. For decades after that race, women were deemed "too fragile" to compete in distance running events. That belief was held so strongly that it took until 1960 for the women's 800m to be reinstated. It took until 1972 before women were allowed to compete in the Olympic 1500m, the same year eight women were "legally" able to run the Boston Marathon for the first time. Fast-forward to 1980 when the American College of Sports Medicine finally came out in defense of women, with a consensus statement advocating for women in competitive running (ACSM 1979). This led to women finally being allowed to compete in the Olympic Marathon for the first time in 1984. Despite these delayed beginnings, women's running has been on the rise, to the point that women now make up over half of all worldwide marathon finishes. However, in contrast to traditional road races, trail and ultrarunning still have a ways to go in closing the gender gap. Women

only represented 23 percent of all ultramarathon finishes worldwide (Knechtle and Nikolaidis 2018). That number is steadily rising, but we still have some catching up to do.

Sadly, exercise physiology research has followed the same disproportionate trends as ultrarunning. Historically, research involving human test subjects has predominantly utilized college-aged males, and that's not just because you can pay college guys next to nothing to do crazy things. The truth is, college-aged males make really convenient volunteers. Females do not, as real and perceived complications stemming from hormonal fluctuations during the menstrual cycle upset the research process. Studies designed with female test subjects generally take twice as long and involve exceptionally precise planning to control for each volunteer and her individual cycle. Throw birth control into the mix and things become even more complicated. All of this costs time, money, and effort, which is not a good excuse, just reality.

This issue is not limited strictly to exercise physiology. The wider scientific community is facing pressure from organizations like the National Institutes of Health to not only include female test subjects in end-stage research but also in pre-clinical trials. So, given the historical lack of research on female athletes, despite the rapid increase in female participation in road running, trail running, and ultrarunning, what do we know about the nuances of female physiology?

Pertinent to the following discussion, we want to set up the differences between the terms “sex” and “gender.” These terms have traditionally, and incorrectly, been used interchangeably. For clarity we will be using the descriptions set forth by the National Institutes of Health, wherein “sex” refers to biological, genetic, cellular, and molecular differences (male vs female), and “gender” refers to the social construct comprised of values inherited by society (men, women, and people who identify as nonbinary).

PHYSIOLOGICAL SIMILARITIES AND DIFFERENCES

Although males and females are similar in many regards, we are also biologically complex. This leads to specific sex differences from the metabolism of certain substrates to your body composition and the fatigability of your muscles. In that vein, it would be naïve not to acknowledge these important anatomical, physiological, and hormonal differences. Some differences might not end up being that impactful on performance outcomes, others may be performance defining, and a few may be just a logical extension of the research (which is my polite way of saying we don't know). We know that endurance performance up to the marathon distance is heavily influenced by factors such as VO₂ max, lactate threshold, and running economy and that these factors explain much of the ~10 percent difference between male and female performances. However, these factors have been shown to be less impactful on ultramarathon performance (Davies et al. 1979). This opens the door for the interplay between other physiological attributes that directly impact performance outcomes, such as substrate utilization and efficiency, fatigue resistance, muscle structure, hormone regulation, gastrointestinal (GI) function, and even decision-making (Knechtle et al. 2018; Millet et al. 2011). But how are these factors swayed by sex, and which ones might be responsible for closing the performance gap between males and females in ultra-endurance events?

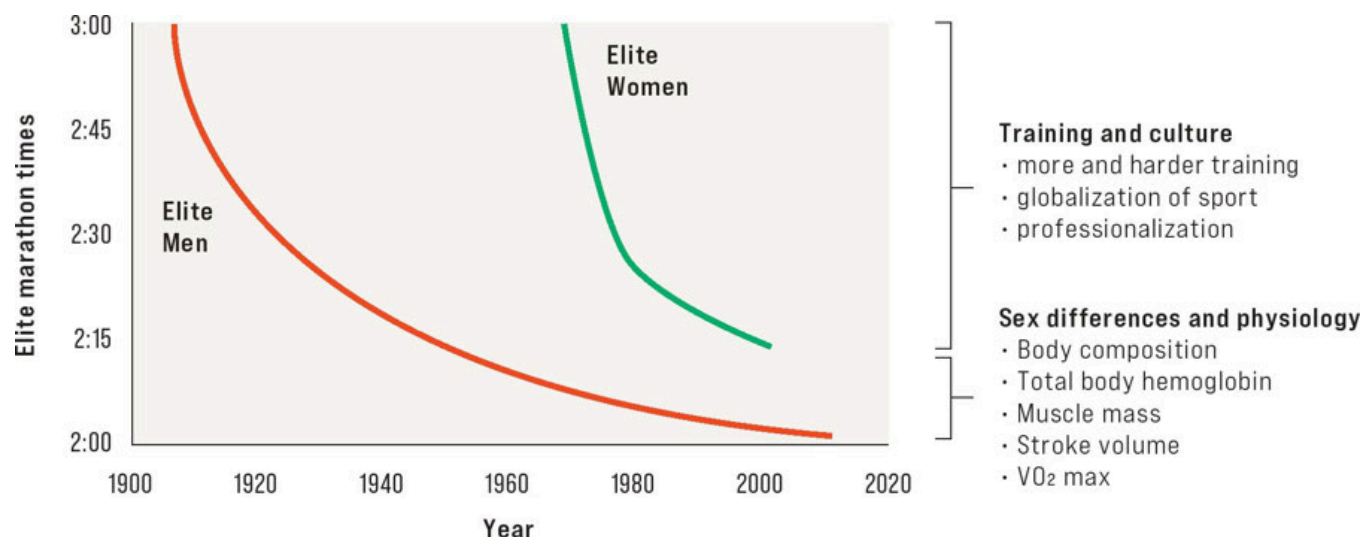


FIGURE 2.3 Many of the initial differences between male and female performance can be attributed to social factors like training and cultural support. The remaining differences that contribute to differences in performance are body composition, total body hemoglobin, muscle mass, stroke volume, and VO_2 max. Source: Joyner 2017.

SUBSTRATE UTILIZATION

We know that carbohydrate and fat provide the majority of the energy required to get down the trail, and this is particularly true during prolonged, submaximal work performed in ultra-endurance events. As races increase in distance and intensity decreases, we rely more and more on the utilization of free fatty acids (fat) to provide energy to working muscles (Waśkiewicz et al. 2012). Doing this can spare our limited store of carbohydrate (glucose stored in muscle and the bloodstream) and stave off the dreaded “bonk” (sudden fatigue due to overdrawing that account). It turns out endurance-trained females exhibit greater rates of fat oxidation (24 to 56 percent more) compared to their male counterparts, even at rest (Melanson et al. 2002). This continues to hold true while exercising. Females tend to utilize more fat as a percentage of total energy expenditure, have higher rates of fat oxidation relative to their body weight, and reach maximal fat oxidation at higher intensities as compared to male athletes (see Figure 2.4; Knechtle et al. 2005; Tarnopolsky et al. 1990; Venables, Achten, and Jeukendrup 2005). Why? Well, it turns out females have at least two things working for them here. The first is that females exhibit greater expression of genes that code for fatty acid metabolism, including a specific protein (CD36) that makes transporting and utilizing free fatty acid more efficient (Kiens et al. 1985; Monaco et al. 2015). Additionally, females have more estrogen than males, which may enhance lipid oxidation and which downregulates glucose utilization (Isacco et al. 2016). It’s these factors, in combination, that seem to give females a distinct metabolic advantage during ultra- and extreme ultra-endurance events. Interestingly, one of the reasons “fat-adapted” strategies are generally ineffective for females is because they are already efficient fat burners (not to mention the potential negative health side effects).

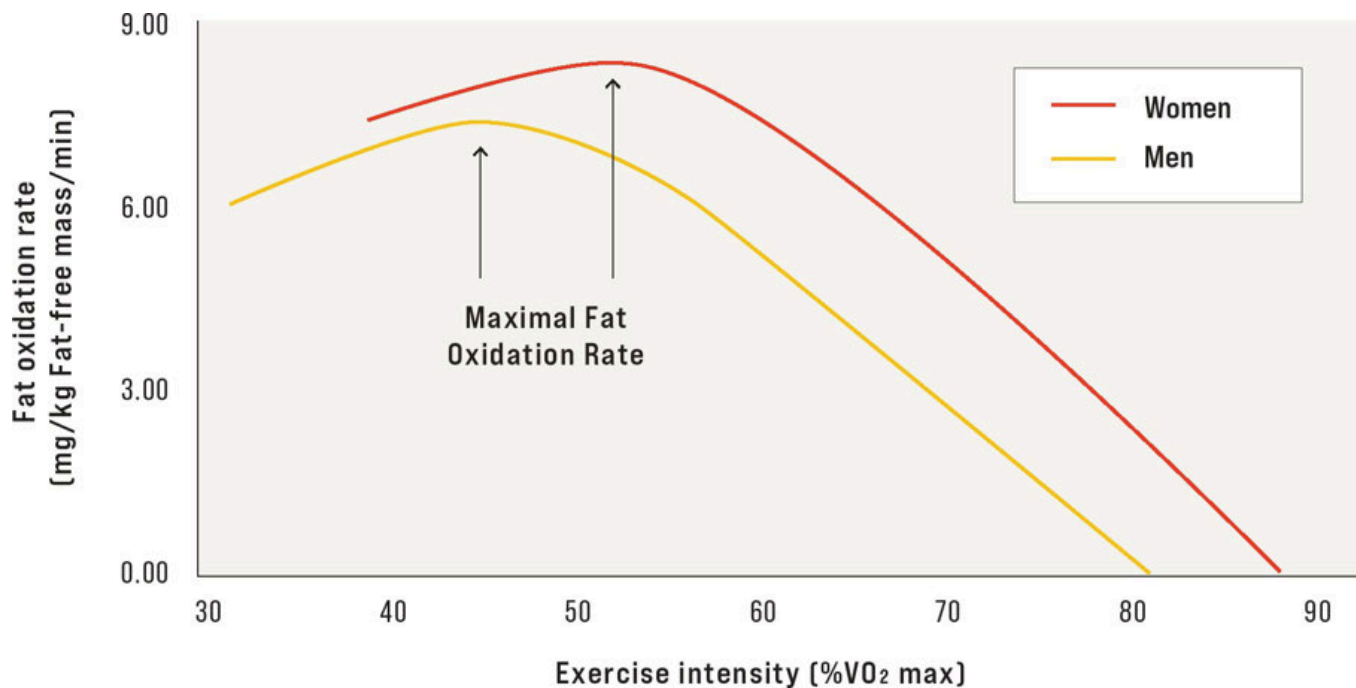


FIGURE 2.4 Women exhibit higher rates of fat oxidation relative to their body mass. Adapted from Venables, Achten, and Jeukendrup 2005.

FATIGABILITY AND MUSCLE STRUCTURE

When it comes to fatigue, in the context of performance limiters, we are specifically addressing neuromuscular fatigue rather than “feeling tired.” Neuromuscular fatigue is what happens when, over the course of a long run, race, or other exhaustive exercise, you experience diminished ability to voluntarily activate and/or contract muscles (Millet 2011). In both the lab and the field, at submaximal intensities, females typically outperform their male counterparts when it comes to muscular fatigability (Hicks et al. 2001; Hunter 2014). At a mechanistic level, there seem to be three factors that are favorable for females in this area: muscle fiber type, muscle mass, and neuromuscular control (Hunter 2014).

You have likely heard about type I and type II muscle fibers (slow-twitch and fast-twitch, respectively), with type II fibers further broken down into oxidative and glycolytic based on their preferred energy pathway (Brooke et al. 1970). Type I fibers are considered more fatigue resistant and are best at long aerobic work due to their high mitochondrial content (Zierath et al. 2004). Interestingly, in a study by Staron et al., they found that females naturally had a higher percentage of type I fibers compared to males (Staron et al. 2000). Additionally, males are more likely to have a higher percentage of glycolytic type II fibers compared to females (Welle et al. 2008). Males are also more likely to have greater muscle mass than their female counterparts, but interestingly this is because their individual muscle fibers are generally larger in diameter (Miller et al. 1993). What this means is that males have less blood flow to their individual muscle fibers during submaximal contractions, making them more fatigable and prone to faster rates of neuromuscular fatigue (Hunter 2014). Combined, these factors create greater demand on the cardiorespiratory and central nervous systems, giving women and their larger percentage of type I muscle fibers a theorized advantage when it comes to long endurance events.

GASTROINTESTINAL DISTRESS

Gastrointestinal (GI) distress is one of the top reasons cited for dropping out of an ultramarathon, next to undertraining (Hoffman et al. 2011). GI distress comes in many flavors, ranging from nausea and vomiting to cramping, reflux, bloating, and diarrhea (Costa et al. 2019; Wilson 2019). Any of these could end your day early from pain or discomfort. As we discuss in Chapter 14, the causes of GI distress during ultramarathons are generally multifactorial and are most commonly exacerbated by taking in more food or fluid than you have practiced, running harder than you have trained for, and failing some element of managing thermoregulation. So, why do females runners report a greater propensity for GI distress?

Two of the main factors that could influence GI distress in females are the physical size of female stomachs and gastric emptying rates. In general, females have 10 percent smaller stomachs than their male counterparts, which would make them less capable of accommodating larger volumes of food (Cox 1945). Additionally, we know that compared to males, females have slower transit times (the time it takes food to go in one end and out the other; Probert et al. 1993) as well as slower gastric emptying rates (how long it takes food to leave the stomach; Mori et al. 2017). The mechanism for this is not exactly understood, but it's thought to be related to concentrations and interactions of estrogen and progesterone (Datz et al. 1987). Essentially, this is one of the logical extensions of the literature: it is not fully understood why females have higher instances of GI distress, but it is apparent that it could be a limiting factor in ultra-endurance performance. The end result is the same, so training the gut is an important factor to consider in your training plan.

HORMONE REGULATION

As mentioned above, males and females have drastically different hormones that regulate their respective reproductive systems. However, those same hormones act on your nonreproductive tissues as well. They play distinct roles in bone and tissue health and even affect exercise performance in various ways (hello muscle mass). These hormones, often referred to as “sex hormones,” are estrogen, progesterone, and testosterone. We'll talk more about estrogen and progesterone and specifically the menstrual cycle in a little bit. One interesting thing to consider is estrogen's role on the elastic properties of muscles and connective tissues (Joyner 2017). These changes are thought to put female athletes at an elevated risk for connective tissue injuries compared to males (The Female ACL 2016), which is something to consider depending on your injury history (e.g., ACL tears). Additionally, it's plausible that running economy might fluctuate for women throughout the course of their menstrual cycle as hormonal levels fluctuate (Casey et al. 2014; Fletcher et al. 2013).

Testosterone has been considered the leading physiological factor in the gap between male and female athletic performances. However, there's a caveat, as testosterone's advantage is seen more in speed and power disciplines compared to endurance sports. This is because testosterone, and exposure to higher circulating levels of it from puberty onward, is responsible for developing greater muscle mass and subsequent strength and power (Storer et al. 2003). It also lowers body fat, altering body composition; creates higher hemoglobin concentrations (Bhasin et al. 2001); and increases protein synthesis to aid in recovery (Griggs et al. 1989). Interestingly, the one area where high testosterone might not be as advantageous is decision-making. Females, on average, are reported to be much better at pacing themselves over the course of ultra-endurance events. This is in no way a slight toward males. I'm blaming the literature here, but higher testosterone levels are generally associated with impulsiveness and greater risk-taking behavior (Goudriaan et al. 2010). Potentially because of this, albeit speculatively, males tend to overestimate their abilities compared to females and therefore have a greater decline in pace over the course of a race (Hubble et al. 2016). Maybe it's testosterone, maybe it's psychology, but that's a topic for another day.

MENSTRUAL CYCLE

It's pretty well established that exercise physiology and performance research focusing on women has not kept pace with the rapid rise of female participation in sport (McNulty et al. 2020). The truth is that science sometimes lags behind what is already being applied in the field. If you are or have been a menstruating female, you can probably self-identify “times of the month” that feel better vs days that feel fairly dreadful. So, what exactly is

going on physiologically during your individual menstrual cycle that might cause you to feel better or worse? And what, if anything, can you do to work with your individual cycle in order to avoid surprises during training or on race day?

The menstrual cycle is a critically important biological rhythm whereby you experience large cyclical fluctuations in endogenous sex hormones, specifically estrogen and progesterone, approximately every twenty-eight days (Mcnulty et al. 2020). These fairly predictable changes in concentrations of estrogen and progesterone across your menstrual cycle create significantly different temporary hormonal phases of your cycle. Although there are distinct phases of every menstrual cycle, typical cycle duration can vary widely between individuals, from twenty-one- to forty-day-long cycles. However, independent of cycle duration, there are three common and sequential phases: the early follicular phase characterized by both low estrogen and progesterone, the ovulatory phase characterized by high estrogen and low progesterone, and the mid-luteal phase characterized by both high estrogen and high progesterone (Mcnulty et al. 2020). Like anything in physiology, I'll say it again and again: it is important to remember that there is a lot of individual variability.

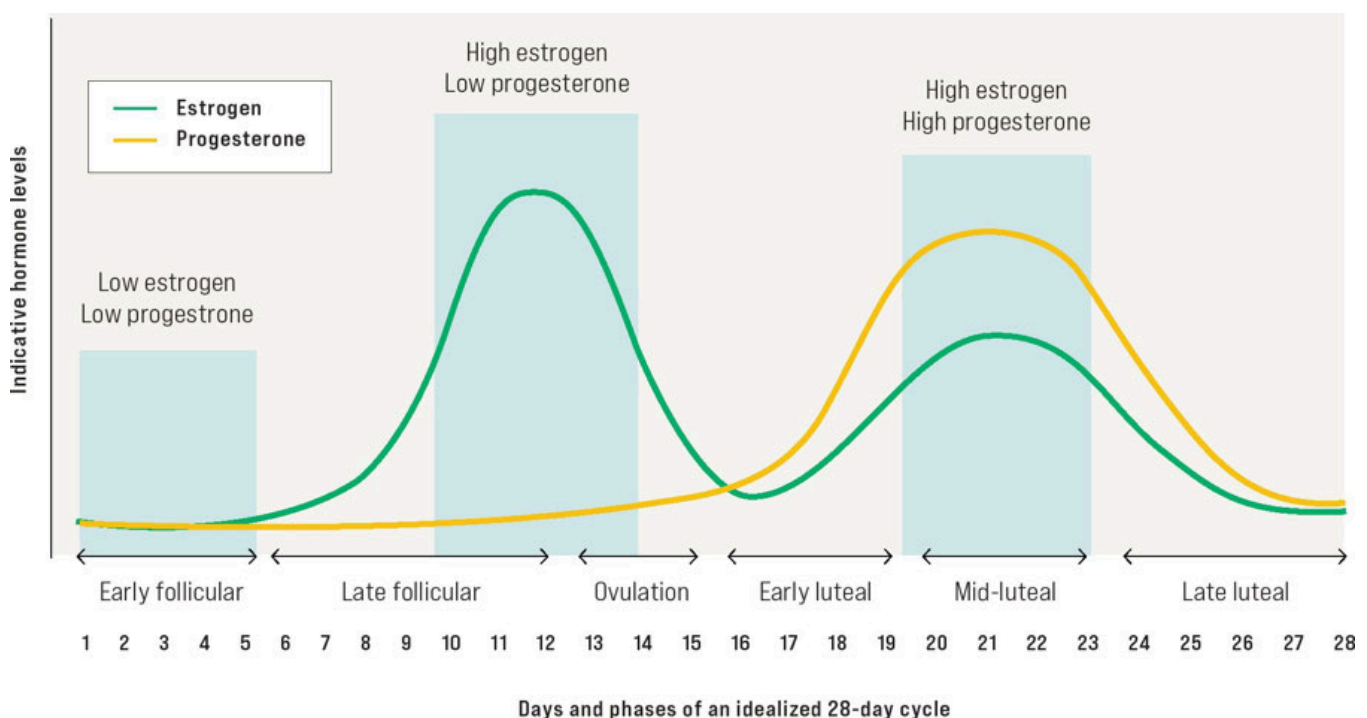


FIGURE 2.5 Representation of the hormonal fluctuation across an idealized twenty-eight-day menstrual cycle.

According to the current literature (which still has a long way to go), menstruating females are most likely to report negative symptoms when estrogen is at its lowest. This time period generally falls within your early follicular phase and means that females often feel their best immediately following this phase, as estrogen rises. Why is estrogen so important? Estrogen has numerous known positive impacts on athletic performance, including regulating glycogen use (and sparing it), protecting against muscle damage and reducing inflammatory response (and thus speeding recovery), and increasing voluntary activation of neural pathways (Mcnulty et al. 2020). Additionally, progesterone is believed to have a dampening effect on estrogen, so as progesterone rises through the luteal phase you might not feel the same estrogen boost as during your follicular phase. Again, for the people in the back, this is based on an idealized cycle; your own individual cycle might create different trends.

We are using a lot of words like, “believed,” “likely,” and “might” to talk about menstrual cycles in relation to athletics, and there are unfortunately two good reasons for that. The first reason is that the research on how the menstrual cycle impacts exercise performance still has too much variance, largely due to methodological issues. The second reason is that, for a long time, coaches and sports scientists have looked at the menstrual cycle in a confined setting and through the lens of a predetermined and static “normal,” when the truth is there is a large

amount of individual variability in both cycle duration and hormonal concentration/phase duration. This means the current recommendation, which comes from a group of researchers far more qualified than me, is to approach training and competing independent of the different phases of your menstrual cycle in as personalized a way as possible (McNulty et al. 2020). From a practical standpoint this means that, as an athlete, you should be tracking your individual menstrual cycle and keeping a log of how you feel and perform at different phases. This should allow you to track trends of when you feel good and when you might experience symptoms and to make individual adjustments to training accordingly. For example, I know that in my early follicular phase I am more likely to experience GI distress. This doesn't mean I can't race during this time, but it does mean I go into training and racing ready to troubleshoot. Over time, as trends may emerge, you can hopefully avoid getting caught off guard by seemingly avoidable surprises.

It's important to acknowledge that this topic area (broadly the topic of females in endurance sport) will need continued research in order for meaningful comparisons to be made, but hopefully this will happen as female participation in ultra-endurance events increases. For now, we can infer that there are several attributes that differ between sexes that might predispose females to find success in ultra-endurance events. Those characteristics include greater fatigue resistance, possessing a greater distribution of type I (oxidative) muscle fibers, a greater efficiency for oxidizing lipids, and the ability to generally pace themselves better than their male counterparts. On the other side of the coin, physiological attributes that might impede female performance include having a higher prevalence of GI distress and the impacts of hormonal fluctuations on metabolism and tissues.

BIRTH CONTROL CONSIDERATIONS

How does birth control impact the menstrual cycle? To start off, there are many different types of birth control, and birth control is used and prescribed for many things aside from contraception. The most common forms of birth control can be divided into two groups: oral contraceptives and intrauterine devices (IUDs). Oral contraceptives are usually taken in a twenty-one-day block followed by a seven-day "contraceptive-free" phase, which allows for a very controlled cycle. This has been touted as a good thing because it means there are generally no surprises. But how does this influence your natural (endogenous) hormones? It turns out intaking exogenous or externally derived estrogens and progestins impacts your natural feedback systems, which results in a chronic downregulation of your hypothalamic-pituitary-ovarian axis (the connection between your hypothalamus, pituitary gland, and ovaries; Elliot-Sale et al. 2020). Basically, it's telling your command center for regulating estrogen and progesterone, "I've got this covered; you can relax." This means your body is no longer in control.

Additionally, one of the biggest differences between a natural cycle and a cycle affected by oral contraceptives is that the daily ingestion of a pill creates a daily surge in hormones, causing mini pseudo-phases. This alteration of your natural hormones has various side effects, but one of them seems to be producing a chronic hormonal profile that is closest to the early follicular phase where both estrogen and progesterone are low. As we suggested earlier, this is generally when females saw the greatest negative impact on their training and racing performance (Elliot-Sale et al. 2020; McNulty et al. 2020). The current consensus in the literature is that being held in that chronic hormonal profile by oral contraception causes slightly impaired exercise performances compared to females who are naturally menstruating. This is consistent with previous studies that suggest that the use of oral contraceptives can lower VO₂ max by ~4.7 percent (Lebrun et al. 2003).

It is important to recognize that there are many times when the best course of treatment for a female athlete may be oral contraceptives. These include: managing the symptoms associated with dysmenorrhea (cramps and pelvic pain) and menorrhagia (heavy bleeding), symptomatic fibroids, functional ovarian cysts and benign breast disease, polycystic ovarian syndrome, premenstrual syndrome, and pelvic inflammatory disease, and decreasing the risk of ovarian and endometrial cancers (Elliot-Sale et al. 2020). On an individual basis, the lessening of those symptoms may far outweigh the small performance decreases associated with taking an oral contraceptive.

The other common contraceptive is IUDs. An IUD can be either hormonal (low-dose progesterone) or nonhormonal (copper) and are popular choices because they are good for three to eight years and do not require remembering to take a daily dose of medication. But wait, didn't we just say that progesterone can blunt the effects of estrogen and mess up our natural hormonal cycles? Generally speaking, yes; however, low-dose

hormonal IUDs keep the hormone localized within your uterus, which means it does not impact your circulating hormone levels the way ingesting an oral contraceptive would.

One risk that comes with either form of contraceptive method is that it can potentially mask amenorrhea. Oral contraceptives can give you a “fake period” while hormonal IUDs cause approximately 20 percent of females to lose the bleeding portion of their menstrual cycle. Both outcomes take away one of our warning signs of poor health and should be taken into account, particularly if you are at risk for relative energy deficiency in sport (RED-S). Finally, if you have any doubt or concern about any aspect of your menstrual cycle or contraceptive options, you should speak with your primary care provider or OB-GYN.

RED-S

I’m not going to mince words—despite many myths, losing your period (the bleeding portion of your menstrual cycle) is not normal, even for endurance athletes. You can think of your period as the canary in the coal mine; if it disappears, that should be a red flag that something is not quite right. Unfortunately, just because you know you shouldn’t lose your period doesn’t mean that it’ll stick around. In fact, amenorrhea (the absence of the menstrual cycle for ninety days or more) and oligomenorrhea (cycles lasting longer than thirty-five days) are reported at rates of up to 70 percent of women participating in sports compared to 2 to 5 percent in the general population (Jacobsson et al. 2014; Nazem et al. 2012).

For a long time amenorrhea, or menstrual dysfunction, was considered one pillar of the Female Athlete Triad, along with low energy availability and decreased bone density. However, more recently the International Olympic Committee and other research groups have coalesced around a similar concept called Relative Energy Deficiency in Sports (RED-S). RED-S still covers the interrelationship between energy availability, reproductive function, and bone health but expands it to acknowledge that males can also fall prey to low energy availability and its associated complications.

Put most simply, your body’s job is to keep you alive, and to do that it needs energy to maintain all your basic physiologic functions, from cellular maintenance and growth to thermoregulation and reproduction. When your body doesn’t receive adequate energy to meet the energetic cost of your exercise, it directs energy away from things like metabolism and reproduction in order to survive. Over time, low energy availability decreases athletic performance and can also impact bone and musculoskeletal health, cardiovascular disease, and neuroendocrine abnormalities like thyroid dysfunction (Clark et al. 2018; Mountjoy et al. 2018). This might sound a little scary, but that is exactly why it is so important. You need to diligently take care of yourself.

It should be noted that you can be in a state of low energy availability and menstrual dysfunction without struggling with restrictive or disordered eating. As endurance athletes, and especially ultra-endurance athletes, it’s easy to fall behind on the nutrition demands of your day-to-day training. This is particularly true because energy expenditure is not static; rather, it’s incredibly dynamic in nature. The ever-changing demands of training mean that you can’t simply balance the calorie checkbook at the end of each calendar day. As a result, you will likely have days of overcompensation, like that on rest day after a long training run over the weekend. Falling into the trap of “limiting your calories” because you are not training that day overlooks the likelihood that you are still behind from a day where your energy expenditure was too much to make up in a twenty-four-hour window.

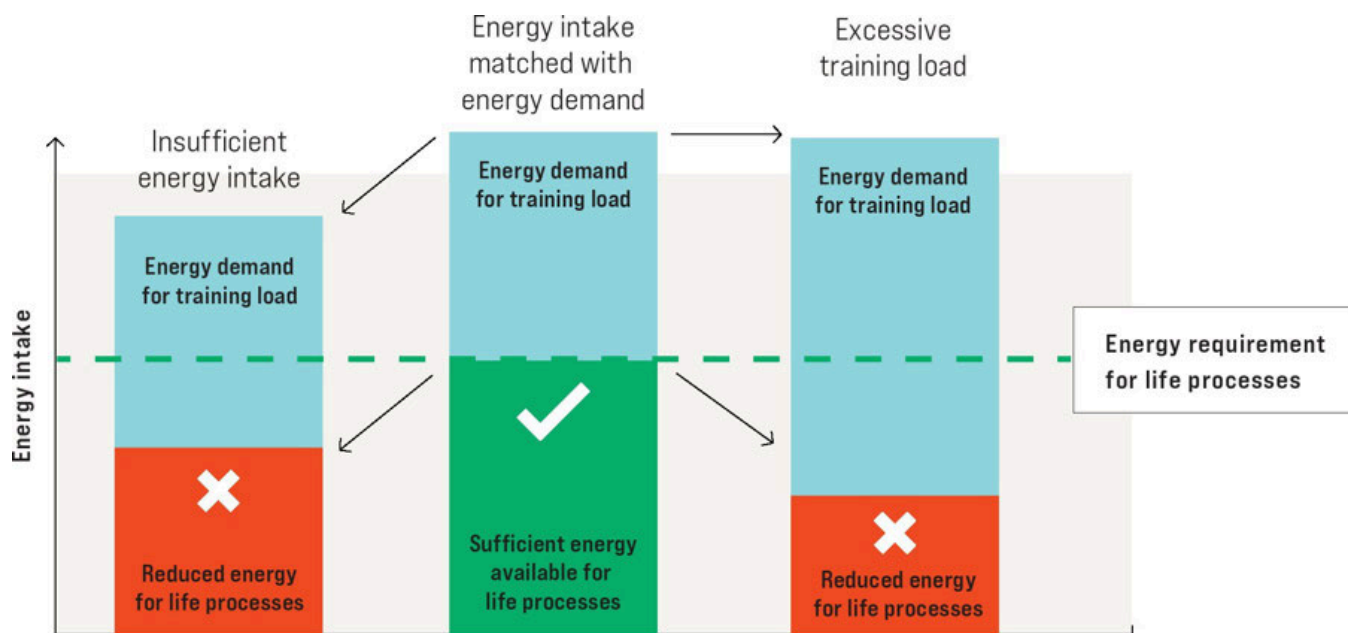


FIGURE 2.6 Matching energy intake to demand. Adapted from Keay 2018.

MENOPAUSE

You’ve hit your stride, and you have time for your ultrarunning pursuits as you revel in the eagerly anticipated “midlife,” but then . . . menopause. It doesn’t seem fair, does it? So, what exactly is going on, and how can you get the best out of your running as the ebb and flow of your hormones changes?

Often striking while the iron is hot in your mid-forties (between 36 and 45 years of age, on average), perimenopause, or “menopause transition,” is the time when your ovaries gradually start to make less and less estrogen. This time period starts on average four to five years before you reach menopause, which is the point at which your ovaries stop releasing eggs and you experience a complete cessation of your period. In the last one to two years leading up to menopause, your drop in estrogen speeds up, and this is when you typically begin to experience the common dreaded menopause-related symptoms like hot flashes; fatigue; mood swings; disrupted sleep; an increase in subcutaneous and intramuscular fat; and, oh yes, performance decline. (Sure, performance declines for males as they age, too, but their falloff point occurs later, as their testosterone concentrations begin to dip.) This decline in female performance is strongly tied to the definitive slowing in the production of key hormones (estrogen, progesterone, and testosterone). During this time there is a gradual acceleration of atrophy and loss of skeletal muscle, predominately type 2 muscle fibers. This loss of lean muscle mass also comes with a reduction in oxygen utilization and a subsequent drop in VO2 max at a rate of 5 to 10 percent per decade after 30 years of age (Hawkins and Wiswell 2003).

So, as you find yourself looking at this seemingly new biological self, what can you do to dampen the blow of a new hormonal profile?

FOCUS ON SLEEP HYGIENE.

As estrogen and progesterone levels decline, your ability to fall asleep and stay asleep declines, too. Combine that with struggles to regulate your body temperature at night due to changes in melatonin and cortisol levels, and

you're exhausted. To counter this, exercise early in the day and do everything you can to practice good sleep hygiene, like sleeping in a cool, dark room away from screens and avoiding alcohol consumption before bed.

LIFT HEAVY THINGS.

I know you are an endurance athlete, but crashing into midlife comes with a gradual and steady loss of lean muscle mass that unfortunately starts sooner for women than for men. This is because that dip in estrogen is accompanied by a dip in testosterone and other growth hormones, making building muscle harder and harder as we age. The best way to counter this and to boost testosterone naturally in the process is to add in ergogenic strength training (see strength training in Chapter 11) combined with adequate protein consumption postexercise. This style of training will also help to maintain healthy bone density.

WATCH OUT FOR THE HEAT.

Hot flashes cause blood to rush to the surface of the skin in an effort to cool you off, which can be annoying when you would prefer your blood to be going to your working muscles or to your stomach to aid in digestion. Additionally, as we age, our sweat rate changes, starting a little later in exercise. This is important because sweating is a primary form of cooling, so slower onset of sweating may lead to increased heat storage at the start of your runs (so a longer warm-up before hard efforts is worth considering). Finally, as we age, our thirst mechanism dulls, making us less in tune with our hydration needs. This means we need to be even more cognizant of our hydration needs during training and racing.

AVOID RUNNING FOR THE BUSHES.

Declining estrogen levels make the body more sensitive to carbohydrates and insulin, while our ability to absorb fructose declines to the point that products with added fructose might send us running to the bushes midrace. This can be managed by avoiding foods with added fructose and looking to sources of energy with simpler sugars such as glucose, dextrose, and sucrose.

DON'T BE AFRAID TO REST.

Recovery takes longer as we age, so don't be afraid to alter your training periodization in both the macro and micro sense. This might mean moving to a ten-day training block instead of following the traditional seven-day week to allow for more rest between hard sessions. Another option is adding in more recovery weeks, thereby moving from a 3:1 or even 4:1 ratio to 2:1.

EAT MORE PROTEIN.

Metabolically, the dramatic slowing and decline in hormone production decreases your insulin sensitivity and your reliance on carbohydrates. This, combined with increased muscle catabolism (the breakdown of muscle tissue), means your ability to maintain lean muscle mass relies heavily on the timing of protein consumption in your post-activity window and, as mentioned above, your ability to lift heavy things.

KEY POINTS ON THE PHYSIOLOGY OF A BETTER ENGINE

You have three pathways with which you can derive energy:

- **The ATP-PCr system**
- **The lactic acid system**
- **The aerobic system**

All three of these are in use all of the time. There are no on-off switches.

There are five basic principles of training:

- **Overload and recovery**
- **Progression**
- **Individuality**
- **Specificity**
- **Systematic Approach**

Females compared to males are predisposed to excel in ultrarunning events in some ways, but not in others.

Female ultrarunners may benefit from tracking their menstrual cycle, along with other variables, and adjusting their training accordingly.

Some women will see performance decline with oral contraceptive use. Use of oral contraceptives should be balanced with the individual goals of each athlete.

Post-menopausal athletes should:

- **Pay extra attention to their sleeping environment**
- **Lift heavy things to avoid loss in muscle mass**
- **Use extra precaution in hot environments**
- **Factor in additional recovery**
- **Pay extra attention to their postexercise and race-day nutrition programs**

